



System Models and Enabling
Technologies

Topic 3



Lecture Outline

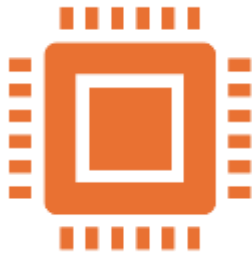
- System Scalability
- Amdahl's Law
- System Availability
- Single Point of Failure
- Security Threats and Defense Technologies
- Types of Attacks and Potential Damages
- System Attacks and Network Threats
- Internet Security Responsibilities
- Case Studies

System Scalability

- The capability of a system to handle an increasing load or demand without performance degradation.
- Importance
 - Critical for system growth
 - Improved performance
 - Long-term sustainability.



Types of Scalability



Vertical Scalability:

Enhancing existing resources, such as adding more CPU, RAM, or storage.



Horizontal Scalability:

Expanding by adding more servers or devices.



Elastic Scalability:

Automatically adjusting resources in response to demand.

Benefits of Scalability



Ensures consistent performance during peak loads.



Reduces system downtime and service disruptions.



Optimizes operational costs by scaling resources as needed.



Supports business growth and user demand.

Amdahl's Law

- Describes the limitations of speedup in parallel computing due to the serial portion of a task.
- Formula: $\text{Speedup} = 1 / (S + (1 - S) / N)$

S = Fraction of the workload that must be executed serially

N = Number of processors



Amdahl's Law

- Scenario: A task where 40% is serial and 60% can be parallelized across 4 processors.
 - $\text{Speedup} = 1 / (0.4 + (0.6/4))$
= 2.22x faster.
- Increasing processors has diminishing returns if the serial portion is significant.

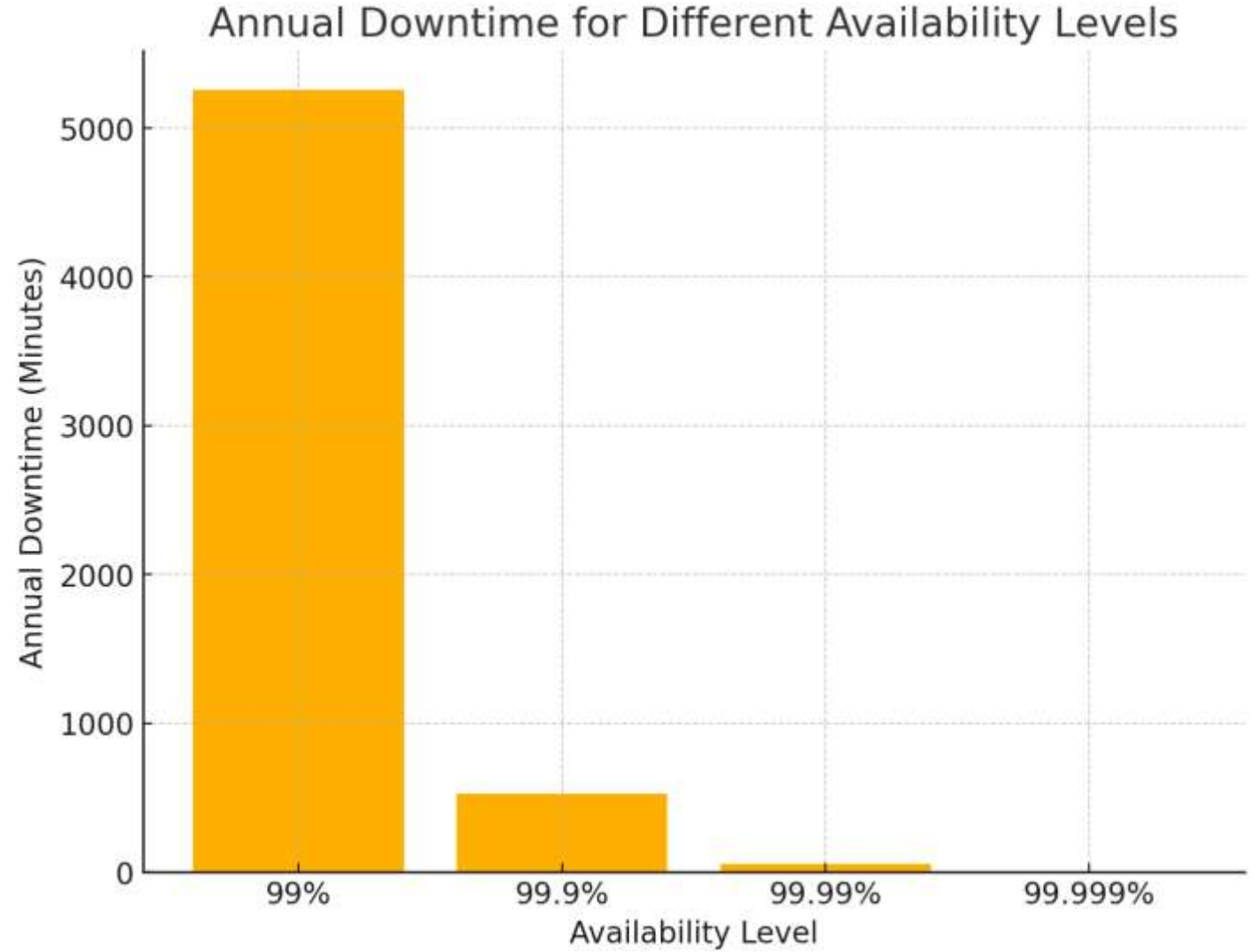


System Availability

- The degree to which a system is accessible and functional when needed.
- Measured as: Availability (%) = (Uptime / Total Time) * 100



Common
Availability
target



Factors Affecting System Availability



Hardware
failures



Software bugs
and crashes



Network
disruptions



Maintenance
and upgrades



Disaster
events
(natural
disasters,
cyberattacks)

Ensuring High Availability



Redundancy: Use backup components and systems.



Load Balancing: Distribute traffic to avoid overloading servers.



Failover Mechanisms: Automatically switch to standby systems during failures.



Monitoring: Continuous monitoring for quick issue detection.

Single Point of Failure (SPOF)

A component whose
failure leads to
total system
shutdown.



Examples:

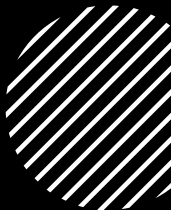
Single
database
server

Central
router in a
network

Power
supply
without
backup



Mitigating SPOF



Use redundant components and backup systems.



Implement clustering and failover solutions.



Distribute services across multiple data centers.

Security Threats and Defense Technologies

Threats: Actions or events that compromise system integrity, confidentiality, or availability.



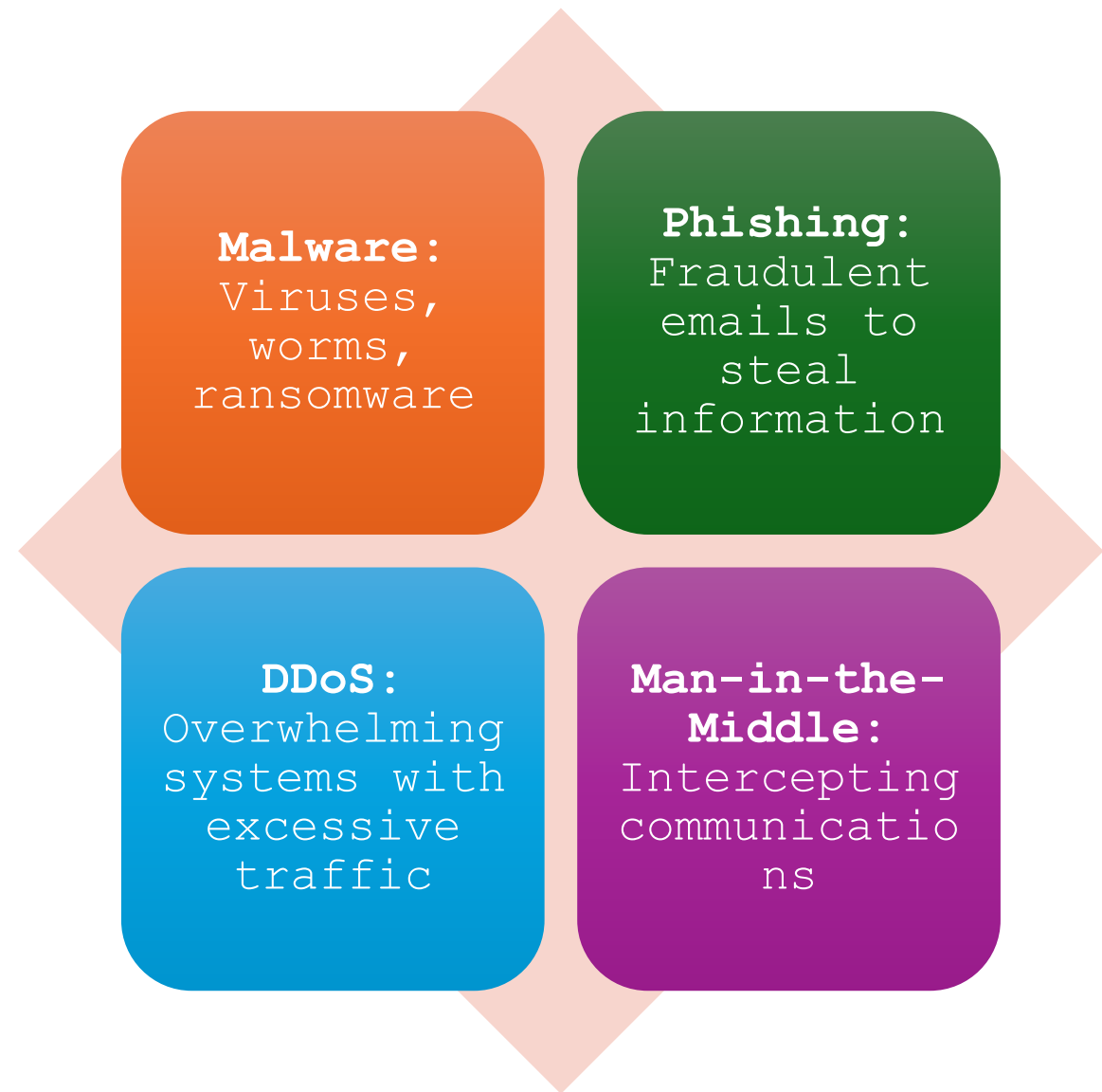
Defense Technologies:

Firewalls
filter network
traffic.

Intrusion
Detection
Systems (IDS)
monitor
suspicious
activity.

Encryption
secures data
during
transmission
and storage.

Types of Security Threats



Potential Damages from Attacks



Data breaches
leading to loss
of sensitive
information



Financial losses
due to business
disruption



Reputational
damage affecting
customer trust



Legal
consequences from
non-compliance

System Attacks



Unauthorized Access: Gaining entry without permission



Privilege Escalation: Increasing access to sensitive areas



Insider Threats: Employees misusing access



Application Vulnerabilities: Exploiting bugs in software

Network Threats to Cyberspac e



PACKET SNIFFING:
CAPTURING DATA PACKETS
DURING TRANSMISSION



IP SPOOFING:
IMPERSONATING TRUSTED
DEVICES



DNS POISONING:
REDIRECTING TRAFFIC TO
MALICIOUS SITES



ZERO-DAY EXPLOITS:
ATTACKS EXPLOITING
UNDISCLOSED
VULNERABILITIES

Defense Against Network Threats



SECURE PROTOCOLS:
USE HTTPS AND VPNS
FOR ENCRYPTION



**NETWORK
SEGMENTATION:**
ISOLATE SENSITIVE
SYSTEMS



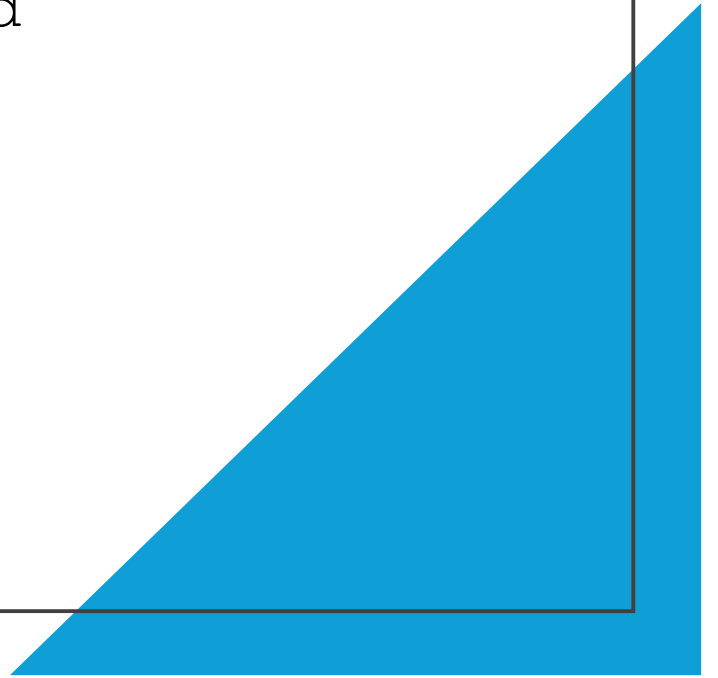
FIREWALLS AND IPS:
BLOCK UNAUTHORIZED
TRAFFIC



PATCH MANAGEMENT:
REGULAR SOFTWARE
UPDATES

Internet Security Responsibil ities

Shared responsibility
between individuals,
organizations, and
governments





Responsibilities of Individuals

- Use strong, unique passwords
- Enable two-factor authentication (2FA)
- Avoid clicking on suspicious links

Responsibilities of Organizations

- Develop comprehensive cybersecurity policies
- Conduct regular employee training
- Use encryption to protect data



Responsibilities of Governments

- Enforce cybersecurity regulations and compliance
- Establish national cybersecurity frameworks
- Collaborate with international cybersecurity agencies



Case Study 1 - Equifax Data Breach

- **Event:** In 2017, hackers stole sensitive information of 147 million individuals.
- **Cause:** Vulnerability in a web application framework
- **Impact:** Massive financial loss, legal actions, and reputational damage
- **Lesson:** Regular vulnerability assessments and timely patching are critical.



Case Study 2 - WannaCry Ransomware Attack

- **Event:** In 2017, a ransomware worm infected systems globally, encrypting data and demanding ransom payments.
- **Cause:** Exploited a vulnerability in outdated Windows systems
- **Impact:** Affected hospitals, businesses, and government agencies worldwide
- **Lesson:** Keep systems updated and use robust backup solutions



Summary

- Scalability and availability are essential for system performance
- Amdahl's Law limits parallel computing benefits
- Security threats must be countered with advanced defense technologies
- Case studies highlight the importance of proactive cybersecurity